The Crypto–FX Channel into Local Equities: Evidence from Major Developed and Emerging Markets

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Abstract

We study the details of how crypto assets truly comove with domestic equities in emerging markets and developed economies. Starting from a tractable portfolio problem in which a representative investor allocates between equities and crypto asset and repatriates wealth each period, we derive a risk-adjusted parity in local-currency real terms. Our model's testable implication is a strong positive co-movement of crypto and equities: domestic equity returns should load positively, including nearly one-for-one, on the FX-converted return of the crypto leg once common shocks are controlled. Using market data, we construct the crypto leg in local currency as the USD crypto return plus the FX change and estimate both in country and a panel data to verify the predictions of our model for crypto. Across all markets, we find comoveements that are strictly positive and has particuarly become very strong in recent years, consistent with our model's variance-bound prediction that equities should link positively with crypto, no more, no less. The results are robust to horizons, other crypto assets in place of BTC, inclusion of controls, and estimation by reverse causation.

1 Introduction

For much of its early history, Bitcoin was heralded as "digital gold"—an uncorrelated asset offering diversification and hedging properties against equities. Portfolio managers and retail investors alike considered crypto an attractive store of value precisely because its returns appeared disconnected from global stock markets. Indeed, before 2020, rolling correlations between Bitcoin and major equity indices hovered near zero, with Box–Whisker plots from 2017 to 2019 showing medians indistinguishable from non-correlation. This narrative shifted dramatically during the COVID-19 crisis and its aftermath. By 2020, correlations between Bitcoin and the S&P 500 or Nasdaq-100 rose sharply, reaching values near 0.5 and remaining persistently positive thereafter. During episodes of heightened stress—the onset of the pandemic in early 2020, the global tightening cycle of 2022, and the volatility of 2023–2025—Bitcoin and equities moved in lockstep, signaling a structural break from the era when crypto traded as an orthogonal risk.

What explains this transition from diversifier to equity-like asset? Several reinforcing factors stand out. First, institutional acceptance has embedded crypto into mainstream portfolios, with asset managers, hedge funds, and even sovereigns (e.g., El Salvador) allocating systematically to digital assets. Second, portfolio integration has deepened comovement, as crypto positions are now rebalanced alongside equities under common risk budgets. Third, volatility dynamics matter: Bitcoin's daily volatility remains higher than that of equities, but the volatility wedge has shrunk significantly over the years, magnifying its role as just another high-beta extension of equity exposure. Fourth, supply and holding patterns have matured, with a rising share of coins absorbed by long-term holders and institutions. Finally, increased access through ETFs, futures, and options has further integrated crypto into the global financial ecosystem. These forces collectively align Bitcoin with the global financial cycle, raising the possibility that crypto no longer hedges equities but instead propagates their risks.

The global trend is particularly consequential for developed markets, several of whom have come to embrace cryptos and emerging markets (EMs). Just as it has gained a great degree of acceptance in the developed markets, cryptocurrency adoption has accelerated most in EMs, where high and volatile inflation, recurrent devaluations, shallow financial intermediation, and capital controls broaden the appeal of dollar-linked stores of value and cross-border payments. In the BRICS and other EMs, for instance, flows into crypto assets have grown rapidly. In the developed markets, there are an increasing number of dedicated funds who main goal is to manage a portfolio of cryptos. Yet the literature offers limited attempts to theoretically explain how crypto markets transmit to domestic asset markets—particularly equities—through the exchange-rate channel, and how this transmission varies across states of the world. For EM investors, crypto assets are not just speculative instruments: because they are denominated in USD, any crypto allocation inherently bundles foreign-exchange exposure with local asset risk. Understanding this linkage is crucial for both policy and portfolio design.

In this paper, we develop and test a model that links three objects central to investors: domestic equity returns, USD crypto returns, and the exchange rate. A representative investor allocates wealth between a local equity and a crypto asset, repatriating proceeds into domestic currency each period. This setup yields a natural parity between local equity returns and a USD-crypto leg converted via the exchange rate. We ask three questions. *First*, does this parity hold empirically in levels and in growth rates, and does it differ across BRICS? *Second*, how does the wedge behave in bad times—when global risk rises, liquidity tightens, or capital controls bind? *Third*, through which channel—crypto innovations or exchange-rate shocks—do disturbances propagate into local equities, and how heterogeneous is this propagation across countries?

Our main contributions are threefold. (1) We formalize a *crypto–FX–equity linkage* and derive propositions that are straightforward to test. (2) We model and measure a state-dependent wedge of crypto-equity returns who volatility has shrunk over the years, offering an explanation for observed comovements. (3) We decompose transmission into *crypto vs. FX channels*, documenting economically meaningful cross-country heterogeneity. These findings are robust to controls and highlight the channels through which crypto exposure permeates EM financial systems.

Our paper connects to several literatures. Early studies characterize the return distribution, volatility, and hedging properties of Bitcoin and other crypto assets (e.g., Dyhrberg, 2016; Baur et al., 2018; Corbet et al., 2018; Urquhart, 2016). Subsequent work explores pricing, factor exposures, and comovement with global risk and equities (Liu and Tsyvinski, 2018; Yermack, 2015; Borri, 2019). On the adoption side, researchers document EM-led uptake and the role of stablecoins, on/off-ramps, and policy frictions. We contribute a portfolio-choice foundation linking USD-crypto to *domestic* equity via FX conversion, and we measure the state-dependent wedge that prior work often treats as residual comovement. Our framework also builds on classic exchange-rate exposure and pass-through studies (Adler and Dumas, 1984; Jorion, 1990), as well as work on global risk and intermediary constraints shaping equity–FX comovement (Hau and Rey, 2006; Bruno and Shin, 2015). The "global financial cycle" literature highlights the dominance of the dollar in transmitting U.S. risk appetite and funding shocks to EM assets; our innovation is to add a crypto twist, quantifying how much of the equity response is attributable to the FX leg versus the USD-crypto innovation itself, and how capital controls and liquidity shape this split.

Our research offers new insights. Relative to existing work, we (i) pose a *crypto–FX–equity* linkage tailored to the broader financial markets, (ii) model a *state-dependent wedge* with concrete empirical implementation, and (iii) provide a *decomposition* of transmission into crypto vs. FX channels in a unified, replication-friendly framework.

The rest of the paper is organized as follows: Section 3.1 presents the model and its implications and develops propositions. Section **??** outlines the empirical strategy. Section **??** reports results and heterogeneity. Section **??** concludes.

2 Stylized Fact for Intuition to Build Motivation

To build intuition for the main idea documented in this paper, we explore how the comovement between stock market returns and Bitcoin returns has evolved since the advent of Bitcoin trading, and how the volatility wedge between these two markets has developed over time. We also examine the wedge between the volatility of Bitcoin returns and equity returns, which provides a sharper view of whether the two asset classes remain distinct or increasingly behave as a single asset class.

As shown in Figure 1, Bitcoin initially exhibited little to no comovement with equities. In fact, the one-year rolling correlation of returns hovered near zero or negative values, supporting the early perception that Bitcoin could act as a hedge against equities. This period coincided with elevated volatility in the returns wedge (the difference between Bitcoin and equity returns) and in the volatility wedge (the difference in volatilities between Bitcoin and equities). In other words, Bitcoin's idiosyncratic risk profile dominated, creating a strong separation from the stock market and reinforcing its appeal as a diversification instrument.

Between 2016 and 2019, the volatility of these wedges began to decline markedly. As wedge volatility compressed, the correlation of Bitcoin with equities rose gradually, with several episodes of positive and statistically significant comovement. This suggests that the distinctiveness of Bitcoin as an independent asset class started to erode as market forces began aligning it more closely with equity performance.

Post-2019, the patterns become striking. The volatility wedges shrank to historically low levels, while the correlation between Bitcoin and equities surged—at times exceeding 0.55 on a one-year rolling basis. Monthly rolling correlations (not shown) occasionally surpass 0.90, indicating that Bitcoin and equity markets are now deeply intertwined. This convergence implies that Bitcoin has lost much of its hedging or diversification property and now often amplifies equity market dynamics rather than offsetting them.

The COVID-19 shock in 2020 and the 2022 monetary tightening cycle provide clear stress tests. In both cases, correlations spiked unusually fast, highlighting Bitcoin's increasing tendency to transmit, rather than diversify, systemic shocks. These crisis dynamics reinforce the broader finding: Bitcoin now behaves more like a high-beta equity factor than as a diversifier.

The evidence points to a shift: Bitcoin's behavior is no longer governed by idiosyncratic drivers alone. As wedge volatility declined, the "idiosyncratic gap" between Bitcoin and equities diminished, and both assets increasingly load on common factors. This convergence is unlikely to be explained by differences in pricing kernels, since both assets are priced under broadly similar stochastic discount factors. Instead, the data suggest that the compression of wedge volatility has eroded Bitcoin's uniqueness as an asset class, making its return dynamics nearly indistinguishable from equities.

From a market perspective, this stylized fact is crucial: the narrative of Bitcoin as "digital gold" or a hedge is increasingly untenable. Instead, Bitcoin's evolution suggests it now acts as a high-beta component of the equity market, at least relative to the past years.

For consistency, we use the highest possible interday frequency—daily returns—and compute one-year rolling windows to smooth short-term noise while retaining the visibility of the documented shifts. Although not reported here, alternative specifications using monthly windows reinforce the same conclusion.

In the next section, to present a model that links bitcoin and equity comovement

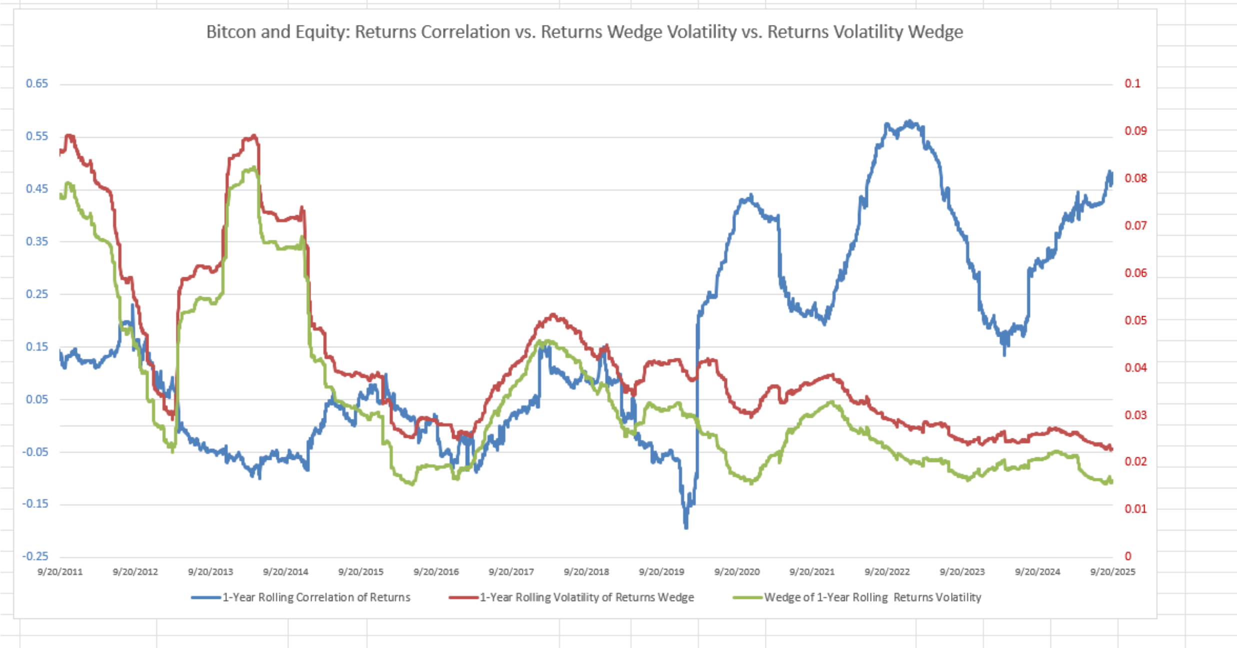

Figure 1: **Bitcoin and Equity:** 1-year rolling correlation of returns (left axis; blue) versus the volatility of the returns wedge (red) and the wedge of 1-year rolling returns volatility (green; right axis). The figure illustrates three phases: an early period of weak/negative comovement with high wedge volatility (2011–2016), a transition with compressing wedges and rising comovement (2016–2019), and a post-2019 convergence where correlations become persistently positive and high. Crisis episodes such as COVID-19 (2020) and the 2022 tightening cycle further reinforced Bitcoin's equity-like behavior.

3 A Model of Equity and Crypto Relationship

Motivated by the reduced-form correlations, the volatility of returns, and the wedge evidence presented in the previous section, we now build a stylized model in which agents hold positions in a domestic equity and in Bitcoin (representative of a USD-denominated crypto asset). Because the investor's wealth is expressed in the domestic currency (LCU), whenever the unit of account differs from the U.S. dollar we must allow the exchange rate to enter the model explicitly. When flows are dollar-denominated or when the domestic currency *is* the dollar, we set the exchange rate to one without loss of generality.

Section 3.1 presents the model setup. Section 3.2 states the associated Bellman equation and solves the first-order conditions. Section 3.3 discusses the propositions implied by the model. Section 3.4 outlines potential extensions.

3.1 Model Setup

To concretize the idea in a tractable environment, consider a representative domestic investor who chooses consumption and portfolio holdings in discrete periods $t = 0, 1, 2, \ldots$. The investor can hold two risky assets: a domestic stock market claim and a global crypto asset (e.g., Bitcoin) that is priced in USD. At the end of each period, all wealth is marked to local currency (LCU). The investor receives exogenous labor income I_t in LCU.

Let C_t denote consumption and P_t^c the domestic price level (CPI, LCU). The gross inflation rate is

$$1 + \pi_{t+1} = \frac{P_{t+1}^c}{P_t^c}.$$

For the domestic stock market, let P_t^s be the stock price (LCU) and D_{t+1} the dividend (LCU) paid at $t+1$.

The gross *nominal* and *real* returns are

$$1 + R^s_{t+1} = \frac{P^s_{t+1} + D_{t+1}}{P^s_t}, \qquad 1 + R^{s,\text{real}}_{t+1} = \frac{1 + R^s_{t+1}}{1 + \pi_{t+1}}.$$

For the cryptocurrency asset priced in USD, let $\Theta^\star_{t+1}$ be the USD return on the crypto asset (e.g., BTC) and define its log return $\theta_{t+1} \simeq \log(1 + \Theta^\star_{t+1})$. Let S_t be the domestic exchange rate expressed as LCU per USD. The FX log return is

$$\Delta s_{t+1} = \log S_{t+1} - \log S_t,$$

where $\Delta s_{t+1} > 0$ means the dollar has appreciated against the LCU.

The investor chooses (i) the number of domestic shares n_t to carry from t to t+1 and (ii) the LCU amount A^k_t to allocate to the crypto leg at t. Investing A^k_t converts to a USD amount A^k_t/S_t at t, earns USD gross return $(1 + \Theta^\star_{t+1})$ over $[t, t{+}1]$, and is repatriated at S_{t+1} into LCU at t+1.

3.2 Budget Constraint

The investor enters period t with positions (n_{t-1}, A^k_{t-1}). After receiving all inflows at t, she chooses (C_t, n_t, A^k_t) and carries only the two risky legs (equity and crypto) into t+1. Thus, in LCU,

Inflows at t:

$$\underbrace{(P^s_t + D_t)\, n_{t-1}}_{\text{stock sale + dividend}} \quad + \quad \underbrace{\frac{S_t}{S_{t-1}}\,(1 + \Theta^\star_t)\, A^k_{t-1}}_{\text{crypto proceeds in LCU after FX}} \quad + \quad \underbrace{I_t}_{\text{labor income}} \quad .$$

Outflows at t:

$$\underbrace{P^c_t C_t}_{\text{consumption}} \quad + \quad \underbrace{P^s_t n_t}_{\text{new stock position}} \quad + \quad \underbrace{A^k_t}_{\text{new crypto allocation}} \quad .$$

Hence the flow budget is

$$P^c_t C_t + P^s_t n_t + A^k_t \;=\; (P^s_t + D_t) n_{t-1} + \frac{S_t}{S_{t-1}}(1 + \Theta^\star_t) A^k_{t-1} + I_t. \tag{1}$$

Define the LCU gross nominal and real returns on the crypto leg:

$$1 + R^k_{t+1} := \frac{S_{t+1}}{S_t}\,(1 + \Theta^\star_{t+1}), \qquad 1 + R^{k,\text{real}}_{t+1} = \frac{1 + R^k_{t+1}}{1 + \pi_{t+1}}. \tag{2}$$

In logs (first-order), $r^k_{t+1} \approx \theta_{t+1} + \Delta s_{t+1}$.

3.3 Investor's Problem and SDF

Following standard arguments, we assume preferences are time-separable with discount factor $0 < \beta < 1$ and strictly increasing, concave $u(\cdot)$. The investor chooses $\{C_t, n_t, A^k_t\}_{t \geq 0}$ to

$$\max\ \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(C_t) \quad \text{s.t.} \quad (1), \quad n_t \geq 0,\ A^k_t \geq 0. \tag{3}$$

Define cash-on-hand (LCU) at time t:

$$M_t := (P^s_t + D_t)\, n_{t-1} \;+\; \frac{S_t}{S_{t-1}}(1 + \Theta^\star_t)\, A^k_{t-1} \;+\; I_t.$$

The investor chooses (C_t, n_t, A^k_t) to solve the Bellman problem

$$V(M_t) \;=\; \max_{C_t, n_t, A^k_t} \Big\{ u(C_t) \;+\; \beta\, \mathbb{E}_t\big[V(M_{t+1})\big] \Big\}$$

subject to

$$P_t^c C_t \;+\; P_t^s n_t \;+\; A_t^k \;=\; M_t, \qquad M_{t+1} \;=\; (P_{t+1}^s + D_{t+1})\, n_t \;+\; \frac{S_{t+1}}{S_t}\,(1+\Theta_{t+1}^{\star})\, A_t^k \;+\; I_{t+1}.$$

Let $V_M(\cdot)$ denote the derivative of V with respect to cash-on-hand. The standard first-order conditions lead to Euler equations in real LCU. Let the real SDF be

$$m_{t+1} := \beta\, \frac{u'(C_{t+1})}{u'(C_t)}\, \frac{1}{1+\pi_{t+1}}.$$

Then

$$\mathbb{E}_t\big[m_{t+1}\,(1+R_{t+1}^s)\big] = 1, \qquad \mathbb{E}_t\big[m_{t+1}\,(1+R_{t+1}^k)\big] = 1,$$

and, equivalently in real terms,

$$\mathbb{E}_t\big[m_{t+1}\,(1+R_{t+1}^{s,\text{real}})\big] = 1, \qquad \mathbb{E}_t\big[m_{t+1}\,(1+R_{t+1}^{k,\text{real}})\big] = 1.$$

Subtracting, we obtain the *risk-adjusted parity*:

$$\boxed{\mathbb{E}_t\Big[m_{t+1}\Big((1+R_{t+1}^s) - (1+R_{t+1}^k)\Big)\Big] = 0} \iff \mathbb{E}_t\big[m_{t+1}\,(R_{t+1}^{s,\text{real}} - R_{t+1}^{k,\text{real}})\big] = 0.$$

Let $w_{t+1} := R_{t+1}^{s,\text{real}} - R_{t+1}^{k,\text{real}}$. From the parity condition we have

$$E_t[w_{t+1}] = -\,\frac{Cov_t(m_{t+1}, w_{t+1})}{E_t[m_{t+1}]}. \tag{2}$$

Decompose w_{t+1} around its conditional mean:

$$w_{t+1} = E_t[w_{t+1}] + \varepsilon_{t+1}, \qquad E_t[\varepsilon_{t+1}] = 0. \tag{3}$$

$$\varepsilon_{t+1} := w_{t+1} - E_t[w_{t+1}]. \tag{3'}$$

Since $R_{t+1}^{s,\text{real}} = R_{t+1}^{k,\text{real}} + w_{t+1}$, it follows that

$$R_{t+1}^{s,\text{real}} = R_{t+1}^{k,\text{real}} - \frac{Cov_t(m_{t+1}, w_{t+1})}{E_t[m_{t+1}]} + \varepsilon_{t+1}, \qquad E_t[\varepsilon_{t+1}] = 0. \tag{4}$$

Corollary Under the model condition, projecting $R_{t+1}^{s,\text{real}}$ on $R_{t+1}^{k,\text{real}}$ yields

$$R_{t+1}^{s,\text{real}} = \alpha + \beta\, R_{t+1}^{k,\text{real}} + u_{t+1}, \qquad E_t[u_{t+1}] = 0,$$

where α captures the average mean adjustment, β measures the strength of comovement, and u_{t+1} is the mean-zero wedge innovation. The model predicts $\beta > 0$ and $\beta \to 1$ as the idiosyncratic wedge variance vanishes.

The structural parity condition implies

$$R_{t+1}^{s,\text{real}} = R_{t+1}^{k,\text{real}} - \frac{Cov_t(m_{t+1}, w_{t+1})}{E_t[m_{t+1}]} + \varepsilon_{t+1}, \qquad E_t[\varepsilon_{t+1}] = 0,$$

so the conditional mean of the wedge is determined by the interaction of the pricing kernel m_{t+1} with the wedge w_{t+1}. In the empirical regression

$$R_{t+1}^{s,\text{real}} = \alpha + \beta R_{t+1}^{k,\text{real}} + u_{t+1},$$

this adjustment cannot be observed directly. Its average effect is absorbed into the intercept α, while any time-varying deviations not captured by α appear in the residual u_{t+1}. Thus α and u_{t+1} together proxy for the SDF–wedge term in empirical implementation.

We now generate the main testable proposition linking the domestic equity and crypto assets. We now state two testable propositions that link the domestic equity leg and the FX-converted crypto leg implied by the first order condition.

3.4 Main Proposition

In the short run, after controlling for common shocks,

$$r_{t+1}^{s,\mathrm{real}} = \alpha + \beta\, r_{t+1}^{k,\mathrm{real}} + u_{t+1}.$$

Let ε_{t+1}^{s} and ε_{t+1}^{k} be the idiosyncratic components of $r_{t+1}^{s,\mathrm{real}}$ and $r_{t+1}^{k,\mathrm{real}}$ after removing common shocks, and define the idiosyncratic parity wedge $e_{t+1} := \varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k}$. Then

$$\beta = \frac{\mathrm{Cov}(\varepsilon_{t+1}^{s}, \varepsilon_{t+1}^{k})}{\mathrm{Var}(\varepsilon_{t+1}^{k})} = 1 + \frac{\mathrm{Cov}(e_{t+1}, \varepsilon_{t+1}^{k})}{\mathrm{Var}(\varepsilon_{t+1}^{k})},$$

and by Cauchy–Schwarz,

$$1 - \sqrt{\frac{\mathrm{Var}(e_{t+1})}{\mathrm{Var}(\varepsilon_{t+1}^{k})}} \;\leq\; \beta \;\leq\; 1 + \sqrt{\frac{\mathrm{Var}(e_{t+1})}{\mathrm{Var}(\varepsilon_{t+1}^{k})}}.$$

In particular, if $\mathrm{Var}(e_{t+1})/\mathrm{Var}(\varepsilon_{t+1}^{k}) < 1$, then $\beta > 0$; moreover, as $\mathrm{Var}(e_{t+1})/\mathrm{Var}(\varepsilon_{t+1}^{k}) \to 0$, we have $\beta \to 1$ as required. [1]

Economic intuition. Both legs are priced by the same SDF; when the idiosyncratic wedge e_{t+1} is small relative to the volatility of the FX-converted crypto leg, the regression slope is forced to lie strictly above zero and concentrate near one. The inequality quantifies this: the tighter the residual wedge, the tighter β is around one. In other words, crypto and domestic equities are positively linked and comove in the short run, rather than offsetting each other. The smaller is the variance of their idiosyncratic wedge, the stronger is the relationship between crypto and equities. This also suggests that periods where equities and crypto returns have offset each other or exhibited very weak links are periods characterized by large wedge variance, driven by factors such as capital controls, liquidity freezes, or policy interventions. Previously, Bitcoin and equities appeared unrelated because the wedge variance was large: thin liquidity, fragmented markets, and regulatory frictions meant crypto returns deviated significantly from equity-linked parity, widening the bounds around the slope β and making estimated correlations indistinguishable from zero. More recently, large institutional, individual, or even sovereign adoption, portfolio integration, and ETF/futures access compressed wedge variance, with the relationship concentrating at higher values and explaining why correlations jumped and stayed persistently positive. Thus, both crypto and equities load on the same stochastic discount factor, and the smaller the wedge variance, the stronger both assets comove and the closer the slope is to one. Our model shows that crypto has therefore shifted from offsetting equity risk to transmitting it—a move from the "digital gold" narrative to equity-like behavior.

3.5 Extensions

The baseline model is intentionally stylized, designed to isolate the core equity–crypto relationship through the FX channel. Nevertheless, several natural extensions could be considered. First, one could introduce heterogeneity in investor types, distinguishing between financially integrated households and liquidity-constrained agents. This would allow wedge variance to depend on market participation and access, potentially explaining cross-country heterogeneity. Second, relaxing the assumption of frictionless markets to incorporate capital controls, liquidity freezes, or transaction costs would make explicit the mechanisms by which wedge variance spikes during stress. Third, the model could be extended to multiple crypto assets (e.g., Bitcoin and Ether), or to stablecoins alongside volatile tokens, to capture a richer menu of USD-linked instruments. Importantly, none of these extensions would overturn the main result of Proposition **??**: as long as both legs load on the same SDF, the short-run slope remains strictly positive and close to unity whenever the idiosyncratic wedge variance is small.

We now move to the empirical analysis section.

[1] Throughout, R denotes a gross return and $r := \log(1 + R)$ its log return. The regression specification is stated in log returns for tractability, but it is directly implied by the structural relation in gross returns.

4 Empirical Analysis

In this section, we now carry out an extensive empirical exercise to examine the linkages between crypto and equities in EM, particularly testing the main implication of the model result that in the short run, domestic real equity returns co-move strongly and positively with the FX-converted real crypto leg (*Proposition* **??**).

4.1 Data and variable construction

Our data includes major EM countries, BRICS countries (Brazil, Russia, India, China, South Africa).

We use the stock market index of each country i and the standard prices of major crypto assets, particularly bitcoin, etherum and...Them we strengthen the validity of our results, we control for Global and EM stress proxies X_t: VIX; EMBI Global; dollar index (DXY); global financial conditions; plus country controls where relevant (capital-control dummies, policy events).

5 Rolling BTC–Equity Correlation (60-day window, Jan 2014–Apr 2025)

. Correlations were near zero before 2020 and turned persistently positive thereafter (0.3–0.6), consistent with Proposition **??** that predicts a strictly positive, near-unity short-run slope once common shocks are controlled.

5.1 Hypothesis

Our main hypothesis is that for each country i, the beta, which measures the reaction of domestic equaties to crypto assets satisfies $\beta_i \approx 1$ or at least is significantly different from zero

$$r^{s,\mathrm{real}}_{i,t+1} = \alpha_i + \beta_i \, r^{k,\mathrm{real}}_{i,t+1} + u_{i,t+1}.$$

5.2 Empirical specifications and inference

For richness of our results, we perform both country-by-country time series empiriical analysis to examine for heterogenieity among countries and we also perform a panel empirical analysis which allows us to take full advantage of the cross sectional and time seriess component of our data. For each country i, We specify our baseline empirical specification as follows:

$$r^{s}_{i,t} = \alpha_i + \beta_i \, r^{k}_{i,t} + u_{i,t}. \tag{4}$$

We then include the relevant vector of control variables X_{it}, yielding

$$r^{s}_{i,t} = \alpha_i + \beta_i \, r^{k}_{i,t} + \gamma_i' X_{it} + u_{i,t}. \tag{5}$$

Panel estimator to allow us to take advantage of the cross-section and time series components:

$$r^{s}_{i,t} = \alpha_i + \beta_i \, r^{k,\mathrm{real}}_{i,t+1} + \gamma_i' X_{it} + \lambda_t + \varepsilon_{i,t}, \tag{6}$$

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Appendix A. Proofs and Further Theoretical Results

This appendix provides proofs for the model implications in Section 3.1 and the formal justification for Propositions **??**–**??**. Throughout, expectations are conditional on $\mathcal{F}_t$ and all returns are *gross* unless otherwise noted. We assume interior solutions ($n_t > 0$, $A_t^k > 0$) and that $\{C_t\}$ is strictly positive and adapted.

A.1. Stochastic discount factor and parity

lemma

[**.** SDF in real LCU and Euler equations] Let m_{t+1} denote the real LCU stochastic discount factor (SDF)

$$m_{t+1} := \beta \frac{u'(C_{t+1})}{u'(C_t)} \frac{P_t^c}{P_{t+1}^c} = \beta \frac{u'(C_{t+1})}{u'(C_t)} \frac{1}{1+\pi_{t+1}}.$$

Then any tradable payoff $1 + R_{t+1}$ quoted in nominal LCU satisfies $_t[m_{t+1}(1 + R_{t+1})] = 1$ and equivalently in real terms $_t[m_{t+1}(1 + R_{t+1}^{\text{real}})] = 1$.

Proof. From the Bellman problem in §3.3, the Lagrange multiplier on the period-t budget constraint is $\lambda_t = u'(C_t)/P_t^c$. FOC for n_t and A_t^k give, respectively,

$$\lambda_t P_t^s = \beta\, _t[\lambda_{t+1}(P_{t+1}^s + D_{t+1})], \qquad \lambda_t = \beta\, _t\left[\lambda_{t+1} \frac{S_{t+1}}{S_t}(1 + \Theta_{t+1}^\star)\right].$$

Divide by λ_t, substitute $\lambda_{t+1}/\lambda_t = (u'(C_{t+1})/u'(C_t))(P_t^c/P_{t+1}^c)$, and rearrange to obtain $_t[m_{t+1}(1+R_{t+1}^s)] = 1$ and $_t[m_{t+1}(1 + R_{t+1}^k)] = 1$. Dividing each return by $1 + \pi_{t+1} = P_{t+1}^c/P_t^c$ gives the real versions. ☐

lemma

[**.** Risk-adjusted parity] For the domestic equity and FX-converted crypto legs in real LCU,

$$E_t\big[m_{t+1}\big((1 + R_{t+1}^{s,\text{real}}) - (1 + R_{t+1}^{k,\text{real}})\big)\big] = 0.$$

Equivalently, letting $w_{t+1} := R_{t+1}^{s,\text{real}} - R_{t+1}^{k,\text{real}}$,

$$E_t[w_{t+1}] = -\frac{Cov_t\big(m_{t+1},\, w_{t+1}\big)}{E_t[m_{t+1}]}\,. \tag{7}$$

A.2. Proof of Proposition ?? (short-run slope $> 0, \approx 1$)

We restate the empirical projection (with time fixed effects as controls for common shocks):

$$r_{t+1}^{s,\text{real}} = \alpha + \beta\, r_{t+1}^{k,\text{real}} + \lambda_t + u_{t+1}, \qquad [r_{t+1}^{k,\text{real}} \lambda_t] = 0,\ [u_{t+1}(\cdot)] = 0.$$

Proof. Let a *single-factor* decomposition capture the shocks common to the two legs:

$$r_{t+1}^{s,\text{real}} = \mu_s + b' F_{t+1} + \varepsilon_{t+1}^{s},$$
$$r_{t+1}^{k,\text{real}} = \mu_k + b' F_{t+1} + \varepsilon_{t+1}^{k},$$

where F_{t+1} stacks common risk shocks (including FX and global conditions) and the idiosyncratic components $\varepsilon_{t+1}^{s}, \varepsilon_{t+1}^{k}$ are mean zero and orthogonal to F_{t+1}. Time fixed effects λ_t span the space of F_{t+1} (up to estimation noise), so the Frisch–Waugh–Lovell theorem implies that the coefficient β in the regression of $r_{t+1}^{s,\text{real}}$ on $r_{t+1}^{k,\text{real}}$ *controlling for* λ_t equals the slope in the auxiliary regression of the residuals:

$$\underbrace{(r_{t+1}^{s,\text{real}} - \mu_s - b' F_{t+1})}_{\varepsilon_{t+1}^{s}} \;=\; \beta\ \underbrace{(r_{t+1}^{k,\text{real}} - \mu_k - b' F_{t+1})}_{\varepsilon_{t+1}^{k}} + u_{t+1}.$$

Hence,

$$\beta = \frac{Cov(\varepsilon_{t+1}^{s}, \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})} = 1 - \frac{Var(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k})}{2Var(\varepsilon_{t+1}^{k})} \;+\; \frac{Var(\varepsilon_{t+1}^{s}) - Var(\varepsilon_{t+1}^{k})}{2Var(\varepsilon_{t+1}^{k})}. \tag{8}$$

$$\beta = \frac{Cov(\varepsilon_{t+1}^{s}, \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})} = 1 + \frac{Cov(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k}, \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})}. \tag{9}$$

By Cauchy–Schwarz,

$$\left|Cov(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k},\, \varepsilon_{t+1}^{k})\right| \;\le\; \sqrt{Var(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k})\, Var(\varepsilon_{t+1}^{k})}.$$

Equivalently, the two-sided bound is

$$-\sqrt{\frac{Var(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})}} \;\le\; \frac{Cov(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k},\, \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})} \;\le\; \sqrt{\frac{Var(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})}}.$$

Thus,

$$1 - \sqrt{\frac{Var(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})}} \;\le\; 1 + \frac{Cov(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k},\, \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})} \;\le\; 1 + \sqrt{\frac{Var(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})}}.$$

Substituting into

$$\beta = 1 + \frac{Cov(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k},\, \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})},$$

we obtain the bound

$$1 - \sqrt{\frac{Var(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})}} \;\le\; \beta \;\le\; 1 + \sqrt{\frac{Var(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})}}.$$

In the absent large state-dependent risk adjustments, we have that $\varepsilon_{t+1}^{s} \approx \varepsilon_{t+1}^{k}$ and $Var(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k})$ is small. Therefore, substitute to obtain

$$\beta \;=\; 1 \;+\; O\!\left(\sqrt{\frac{Var(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k})}{Var(\varepsilon_{t+1}^{k})}}\right),$$

so $\beta \approx> 1$ and, in particular, $\beta > 0$ whenever $Var(\varepsilon_{t+1}^{s} - \varepsilon_{t+1}^{k}) < Var(\varepsilon_{t+1}^{k})$.

or

$$\beta \;=\; 1 \;-\; O\!\left(\sqrt{\frac{Var(\varepsilon^s_{t+1} - \varepsilon^k_{t+1})}{Var(\varepsilon^k_{t+1})}}\right),$$

so $\beta \approx> 1$ and, in particular, $\beta > 0$ whenever $Var(\varepsilon^s_{t+1} - \varepsilon^k_{t+1}) < Var(\varepsilon^k_{t+1})$.

where

$$O\!\left(\sqrt{\frac{Var(\varepsilon^s_{t+1} - \varepsilon^k_{t+1})}{Var(\varepsilon^k_{t+1})}}\right) \;\to\; 0 \quad \text{as} \quad \left(\sqrt{\frac{Var(\varepsilon^s_{t+1} - \varepsilon^k_{t+1})}{Var(\varepsilon^k_{t+1})}}\right), \;\to\; 0,$$

because $Var(\varepsilon^s_{t+1} - \varepsilon^k_{t+1}) \to 0$.

Therefore $\beta > 0$ and $\beta \approx> 1$ whenever the idiosyncratic discrepancy between the two legs is second-order relative to the variation in the crypto leg after removing common shocks. $\square$